

User_Guide

WebSocket Monitoring System - User Guide

Getting Started

This guide helps you monitor your translation workflows in real-time using WebSocket-based monitoring.

What You Can Monitor

- **Translation Progress:** See real-time progress updates as your documents are translated
- **SSH Workers:** Monitor remote translation workers and their performance
- **Error Tracking:** Get immediate alerts when errors occur
- **Session History:** View past translation sessions and their results
- **Performance Metrics:** Track translation speed and worker efficiency

Step-by-Step Instructions

Step 1: Start the Monitoring Server

1. Open your terminal
2. Navigate to your project directory:

```
cd /Users/milosvasic/Projects/Translate
```

3. Start the monitoring server:

```
go run ./cmd/monitor-server
```

4. You should see output like:

```
🚀 Starting Translation Monitoring Server
📡 WebSocket server listening on ws://localhost:8090/ws
🌐 Web dashboard available at http://localhost:8090/monitor
```

Keep this terminal open - the monitoring server needs to keep running.

Step 2: Open the Monitoring Dashboard

1. Open your web browser
2. Go to: `http://localhost:8090/monitor`
3. You should see a monitoring dashboard with:
 - Active sessions counter
 - Connection status indicator
 - Event log area
 - Progress tracking section

For advanced features, you can also open: `file:///Users/milosvasic/Projects/Translate/enhanced-monitor.html`

Step 3: Run a Translation with Monitoring

Choose one of the following options:

Option A: Quick Demo (Easiest)

1. Open a **new terminal window**
2. Run the basic demo:

```
cd /Users/milosvasic/Projects/Translate
go run demo-translation-with-monitoring-fixed.go
```

3. Watch the dashboard update in real-time!

Option B: Real AI Translation

1. **Optional:** Set up your OpenAI API key:

```
export OPENAI_API_KEY=your_api_key_here
```

2. Run the real LLM demo:

```
go run demo-real-llm-with-monitoring.go
```

Option C: SSH Worker Translation

1. **Optional:** Configure SSH worker (if you have remote workers):

```
export SSH_WORKER_HOST=localhost
export SSH_WORKER_USER=milosvasic
```

2. Run the SSH worker demo:

```
go run demo-ssh-worker-with-monitoring.go
```

Step 4: Monitor Your Translation

As your translation runs, you'll see in the dashboard:

Real-time Updates

- **Progress Bar:** Shows overall completion percentage
- **Current Step:** What's happening right now (reading, translating, etc.)
- **Event Log:** Detailed log of all activities
- **Session ID:** Unique identifier for your translation session

What the Progress Means

- **0-10%:** Reading and parsing input files
- **10-25%:** Content preparation and conversion
- **25-85%:** Actual translation work
- **85-100%:** Generating output files

Color-Coded Status

-  **Yellow:** Starting up or in progress
-  **Green:** Successfully completed
-  **Red:** Error occurred
-  **Gray:** Pending or idle

Step 5: Understanding the Results

When Translation Completes Successfully

You'll see:

-  Green "Completed" status
- 100% progress bar
- Output file location
- Total time taken
- Any worker information (for SSH translations)

If Errors Occur

You'll see:

-  Red error status
- Detailed error message
- Step where error occurred
- Troubleshooting suggestions

Advanced Features

Monitoring Multiple Sessions

1. Run multiple translations in parallel:

```
# Terminal 1
```

```
go run demo-translation-with-monitoring-fixed.go
```

```
# Terminal 2 (simultaneously)
go run demo-translation-with-monitoring-fixed.go

# Terminal 3 (simultaneously)
go run demo-translation-with-monitoring-fixed.go
```

2. The dashboard will show all active sessions

SSH Worker Monitoring

If you have remote SSH workers configured:

- You'll see worker host and connection info
- Worker model and capacity details
- Real-time worker performance
- Connection status for each worker

Custom Session Monitoring

1. In the dashboard, enter a specific session ID
2. Click "Monitor Session" to focus on one translation
3. View detailed progress and events for that session

Troubleshooting Common Issues

"Cannot connect to WebSocket server"

Solution: Make sure the monitoring server is running:

```
# Check if server is running
lsof -i :8090

# If not running, start it:
go run ./cmd/monitor-server
```

"No events appearing in dashboard"

Solutions:

1. Check that a translation is actually running
2. Verify the translation process includes monitoring code
3. Check browser console for JavaScript errors

"SSH worker connection failed"

Solutions:

1. Verify SSH credentials are correct
2. Test SSH connection manually:

```
ssh milosvasic@localhost 'echo "SSH works"'
```

3. Check worker configuration in config files

"LLM API authentication failed"

Solutions:

1. Verify your API key is set:

```
echo $OPENAI_API_KEY
```

2. Set the API key:

```
export OPENAI_API_KEY=your_actual_api_key
```

"Port already in use"

Solution: Kill the process using port 8090:

```
# Find the process
```

```
lsof -i :8090
```

```
# Kill it (replace PID with actual process ID)
```

```
kill -9 <PID>
```

```
# Restart server
```

```
go run ./cmd/monitor-server
```

Performance Tips

For Best Experience

- Use a modern web browser (Chrome, Firefox, Safari)
- Keep the monitoring server on a stable network
- Don't run too many translations simultaneously on a slow computer

Monitoring Multiple Workers

- Use the enhanced dashboard for better multi-worker support
- Consider increasing WebSocket timeout for slow connections
- Monitor worker capacity to avoid overloading

Large Translation Files

- Progress updates may be less frequent for large files
- Monitor memory usage on the translation machine
- Consider breaking very large files into smaller chunks

FAQ

Q: Do I need to keep the monitoring server running? A: Yes, the monitoring server must stay running to track translations and serve the dashboard.

Q: Can I monitor translations running on different machines? A: Yes! Configure the translation clients to connect to your monitoring server's IP address instead of localhost.

Q: How long does the monitoring history last? A: Session history is stored in memory and lasts until you restart the monitoring server.

Q: Is my data secure? A: For development, data is sent over plain WebSocket. In production, consider using WSS (WebSocket Secure) and authentication.

Q: Can I export the monitoring data? A: Currently, data is displayed in the dashboard only. Export functionality is planned for future versions.

Getting Help

If you encounter issues:

1. **Check the logs** in your terminal windows
2. **Look at browser console** (F12 → Console tab)
3. **Try the basic demo first** to ensure the system works
4. **Review the technical documentation** in docs/
WebSocket_Monitoring_Guide.md
5. **Check configuration files** for any incorrect settings

Keyboard Shortcuts (Dashboard)

- **Ctrl+R**: Refresh dashboard
- **Ctrl+L**: Clear event log
- **F5**: Full page refresh

Quick Reference

Command	Purpose
go run ./cmd/monitor-server	Start monitoring server
go run demo-translation-with-monitoring-fixed.go	Run basic demo
go run demo-real-llm-with-monitoring.go	Run real LLM demo
go run demo-ssh-worker-with-monitoring.go	Run SSH worker demo
http://localhost:8090/monitor	Basic dashboard
enhanced-monitor.html	Advanced dashboard

Success Checklist

Before you start monitoring real translations:

☐

Monitoring server running without errors

☐

Dashboard accessible in browser

☐

At least one demo runs successfully

☐

Real-time updates appear in dashboard

☐

No connection errors in browser console

☐

Translation output files are created

When all items are checked, you're ready to monitor your actual translation workflows!

Happy Monitoring! 🚀